

Assem Moussa Badr

AI Engineer

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Professional Summary

AI Engineer with hands-on experience in machine learning, deep learning, and speech recognition systems. Experienced in fine-tuning Transformer-based models and deploying real-time AI applications using Microsoft Azure. Passionate about building scalable AI systems and solving real-world problems.

Education

Bachelor's Degree in Computer and Information Science, Ain Shams University. (2020 – 2024)

Grade: Very Good.

Experience

- Built and fine-tuned Transformer-based speech recognition models (Whisper).
- Developed real-time AI systems using WebSockets and deployed them on Microsoft Azure.
- Implemented data preprocessing pipelines and model evaluation techniques.
- Worked on end-to-end AI systems from training to deployment.

Technical Skills

Programming: Python, C++, C#, Java, JavaScript, Scala.

Machine Learning: Neural Networks, Model Evaluation.

Deep Learning: Transformers, PyTorch, NLP, Speech Recognition.

Data & Tools: Pandas, NumPy, OpenCV.

Cloud & Deployment: Microsoft Azure, WebSockets.

Concepts: Data Structures, Algorithms, OOP.

Projects

Makhraj – Automatic Speech Recognition System (Graduation Project, Grade: A)

- Applied audio preprocessing (MFCC).
- Fine-tuned Transformer (Whisper) model using PyTorch.
- Provided real-time feedback using WebSockets.
- Deployed system on Microsoft Azure.

Dry Bean Classification

- Conducted comparative analysis between Perceptron and Adaline for binary classification, optimizing performance through feature selection.
- Applied core machine learning concepts, analyzing differences between gradient-based and threshold-based learning approaches.

Mini Amazon

- Developed e-commerce system using C++ and Data Structures.

Blackjack Game

- Java-based game using OOP principles.

Languages

- Arabic (Native)
- English (B2)